
PHYSICS

9702/41

Paper 4 A Level Structured Questions

October/November 2019

MARK SCHEME

Maximum Mark: 100

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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This document consists of **15** printed pages.



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PUBLISHED**Generic Marking Principles**

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptors for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Question	Answer	Marks
1(a)	force proportional to product of masses and inversely proportional to square of separation	B1
	idea of (gravitational) force between point masses	B1
1(b)(i)	above the equator	B1
	from west to east	B1
1(b)(ii)	gravitational force provides/is the centripetal force	B1
	$GM/r^2 = r(2\pi/T)^2$	C1
	$(6.67 \times 10^{-11} \times M) = \{(4.23 \times 10^7)^3 \times 4\pi^2\} / (24 \times 3600)^2$	C1
	$M = 6.0 \times 10^{24} \text{ kg}$	A1

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Question	Answer	Marks
2(a)	(total volume of molecules is) negligible	M1
	compared with volume occupied by the gas	A1
2(b)(i)	$pV = NkT$	C1
	$4.60 \times 10^5 \times 2.40 \times 10^{-2} = N \times 1.38 \times 10^{-23} \times (273 + 23)$	C1
	or	
	$pV = nRT$	(C1)
	$4.60 \times 10^5 \times 2.40 \times 10^{-2} = n \times 8.31 \times (273 + 23)$ $n = 4.49 \text{ (mol)}$ $N = nN_A$ $= 4.49 \times 6.02 \times 10^{23}$	(C1)
	$N = 2.7 \times 10^{24}$	A1

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Question	Answer	Marks
2(b)(ii)	volume of one atom = d^3 ($= 2.7 \times 10^{-29} \text{ m}^3$)	C1
	volume of all atoms = $2.7 \times 10^{-29} \times 2.7 \times 10^{24}$	C1
	$= 7 \times 10^{-5} \text{ m}^3$	A1
	or	
	volume of one atom = $(4/3)\pi r^3$ ($= 1.41 \times 10^{-29} \text{ m}^3$)	(C1)
	volume of all atoms = $2.7 \times 10^{24} \times 1.41 \times 10^{-29}$	(C1)
	$= 4 \times 10^{-5} \text{ m}^3$	(A1)
2(c)	numerical comparison between answer to (b)(ii) and $2.4 \times 10^{-2} \text{ (m}^3\text{)}$ showing (b)(ii) is <u>much</u> less than $2.4 \times 10^{-2} \text{ (m}^3\text{)}$	B1

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Question	Answer	Marks
3(a)	(thermal) energy per (unit) mass (to change state)	B1
	(heat transfer during) change of state at constant temperature	B1
3(b)(i)	32 g	A1
3(b)(ii)	temperature difference (between liquid and surroundings) does not change	B1
3(b)(iii)	$VIt = mL$	C1
	$230 \times 1.2 \times 60 \times 10 = (56 \times L) + H$ or $190 \times 1.0 \times 60 \times 10 = (32 \times L) + H$	C1
	$86 \times 600 = (56 - 32) \times L$	C1
	or	
	$230 \times 1.2 = (56 \times L) / (60 \times 10) + P$ or $190 \times 1.0 = (32 \times L) / (60 \times 10) + P$	(C1)
	$276 - 190 = (24 \times L) / 600$	(C1)
	$L = 2200 \text{ J g}^{-1}$	A1

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Question	Answer	Marks
3(b)(iv)	$230 \times 1.2 \times 600 = (56 \times 2150) + H$ or $190 \times 1.0 \times 600 = (32 \times 2150) + H$	C1
	$H = 45\,200$ rate = $45\,200 / 600$ = 75 W	A1
	or	
	$230 \times 1.2 = (56 \times 2150) / (60 \times 10) + P$ or $190 \times 1.0 = (32 \times 2150) / (60 \times 10) + P$	(C1)
	rate (= P) = 75 W	(A1)

Question	Answer	Marks
4(a)(i)	distance from a (reference) point in a given direction	B1
4(a)(ii)	line is not straight or gradient is not constant	B1
4(b)(i)	0.85–0.90 cm	A1
4(b)(ii)	$a = -(2\pi f)^2 x$	C1
	e.g. $1.2 = 4\pi^2 \times f^2 \times (0.90 \times 10^{-2})$	C1
	$f = 1.8\text{ Hz}$	A1
4(c)	complete circle/ellipse enclosing the origin	B1
	closed shape passing through $(0, \pm v_0)$ and $(\pm x_0, 0)$	B1

Question	Answer	Marks
5(a)(i)	provides return for the signal	B1
	shields signal from noise	B1
5(a)(ii)	e.g. connection between aerial and TV set	B1
5(b)(i)	gain / dB = $10 \lg (P_1 / P_2)$	C1
	$32 = 10 \lg \{P_{\text{MIN}} / (7.6 \times 10^{-6})\}$	A1
	$P_{\text{MIN}} = 0.012 \text{ W}$	
5(b)(ii)	attenuation per unit length = $(1 / L) \times 10 \lg (P_1 / P_2)$	C1
	$6.3 = (1 / L) \times 10 \lg (2.6 / 0.012)$	
	$L = 3.7 \text{ km}$	A1

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Question	Answer	Marks
6(a)	$(E =) Q / 4\pi\epsilon_0 r^2$	M1
	where ϵ_0 is permittivity (of free space)	A1
6(b)(i)	field does not change direction/field does not become zero	M1
	so (charges have) opposite (sign)	A1
6(b)(ii)	minimum is at the midpoint (between the charges)	M1
	so (magnitudes are the) same	A1
6(c)	force = field strength $\times$ charge and force = mass $\times$ acceleration or acceleration is proportional to field strength	B1
	(from $x = 3.0$ cm) to $x = 5.0$ cm: acceleration decreases	B1
	at $x = 5.0$ cm: acceleration is a minimum	B1
	from $x = 5.0$ cm (to $x = 7.0$ cm): acceleration increases	B1

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Question	Answer	Marks
7(a)(i)	all frequencies are amplified equally	B1
7(a)(ii)	no drop in output voltage (when there is a current)	B1
7(b)(i)	gain = $1 + R_F / R_{IN}$ $= 1 + (4000 / 800)$	C1
	$= 6.0$	A1
7(b)(ii)	$2.0 / V_{IN} = 6.0$ $V_{IN} = (+)0.33 \text{ V}$	A1
7(b)(iii)	5.0 V	A1
7(b)(iv)	$V = 5.0 - 2.2 (= 2.8 \text{ V})$	C1
	$R = V / I$ $= 2.8 / 0.020$ $= 140 \Omega$	A1

Question	Answer	Marks
8(a)	concentric circles (around the wire)	M1
	at least 3 circles shown, all with increasing separation	A1
	direction anticlockwise	B1
8(b)(i)	$B = (4\pi \times 10^{-7} \times 6.2) / (2\pi \times 3.1 \times 10^{-2})$	C1
	$= 4.0 \times 10^{-5} \text{ T}$	A1
8(b)(ii)	$F = BIL$ or $F/L = BI$	C1
	$F/L = 4.0 \times 10^{-5} \times 8.5$	A1
	$= 3.4 \times 10^{-4} \text{ N m}^{-1}$	
8(c)	correct application of Newton's 3rd law to the forces or F/L is proportional to the product of the two currents	M1
	so same magnitude	A1

Question	Answer	Marks
9(a)	nuclei precess	B1
	precession is about (direction of magnetic) field or frequency of precession is in radio-frequency range	B1
9(b)	- frequency (of precession) depends on field strength - to locate/find position of (spinning) nuclei - to change region where nuclei are detected any two points, one mark each	B2

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Question	Answer	Marks
10(a)(i)	lower right and upper left diodes circled	B1
10(a)(ii)	maximum = $7.0\sqrt{2}$ = 9.9 V	A1
10(b)(i)	correct symbol for capacitor, shown connected in parallel with R	B1
10(b)(ii)	1. (ripple) decreases	B1
	2. (ripple) increases	B1

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Question	Answer	Marks				
11(a)	energy (of photon) required to remove electron	M1				
	from a surface or reference to <u>minimum</u> energy or reference to zero <u>kinetic</u> energy	A1				
11(b)(i)	1. photon energy = hc / λ	C1				
	$= (6.63 \times 10^{-34} \times 3.00 \times 10^8) / (280 \times 10^{-9})$ $= 7.1 \times 10^{-19} \text{ J}$	A1				
	2. electron energy = $(7.1 - 5.5) \times 10^{-19} \text{ J}$	C1				
	$\frac{1}{2} \times 9.11 \times 10^{-31} \times v^2 = (7.1 - 5.5) \times 10^{-19}$	C1				
	$v = 5.9 \times 10^5 \text{ m s}^{-1}$	A1				
11(b)(ii)	energy is required to bring electron to the surface	B1				
11(c)	<table><tr><td>no change</td><td>decreases</td></tr><tr><td>increases</td><td>decreases</td></tr></table>	no change	decreases	increases	decreases	B4
no change	decreases					
increases	decreases					

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Question	Answer	Marks
12(a)	$E = (3.0 \times 10^8)^2 \times 1.66 \times 10^{-27} (= 1.49 \times 10^{-10} \text{ J})$	M1
	$= (1.49 \times 10^{-10}) / (1.60 \times 10^{-19}) = 9.34 \times 10^8 = 934 \text{ MeV}$	A1
	or	
	binding energy = 8.443×95 [or equivalent using La-139 nucleus]	(M1)
	binding energy / mass defect = $(8.443 \times 95) / 0.859 = 934 \text{ MeV}$	(A1)
12(b)(i)	binding energy = $1.865 \times 934 (= 1741.91 \text{ MeV})$	C1
	binding energy per nucleon = $1741.91 / 235$ $= 7.41 \text{ (MeV)}$	A1
12(b)(ii)	less (than)	B1
12(c)	energy = $\{(1.219 + 0.859) - 1.865\} \times 934$ or energy = $(95 \times 8.443) + (139 \times 8.189) - (235 \times 7.412)$	C1
	$= 199 \text{ MeV}$	A1
12(d)	number of reactions = $1.2 \times 10^{-7} \times 6.02 \times 10^{23}$ $= 7.22 \times 10^{16}$	C1
	energy release (for one reaction) = $199 \times 1.60 \times 10^{-13} (= 3.18 \times 10^{-11} \text{ J})$	C1
	power = $(7.22 \times 10^{16} \times 3.18 \times 10^{-11}) / (25 \times 10^{-3})$ $= 9.2 \times 10^7 \text{ W}$	A1